MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, MODULAR, TYPE IV, CONNECTOR, 100 PIN, CENTER, CROSSOVER, CONTACT TAILS ON .050 CENTERS (CERAMIC COMPATIBLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-28754.

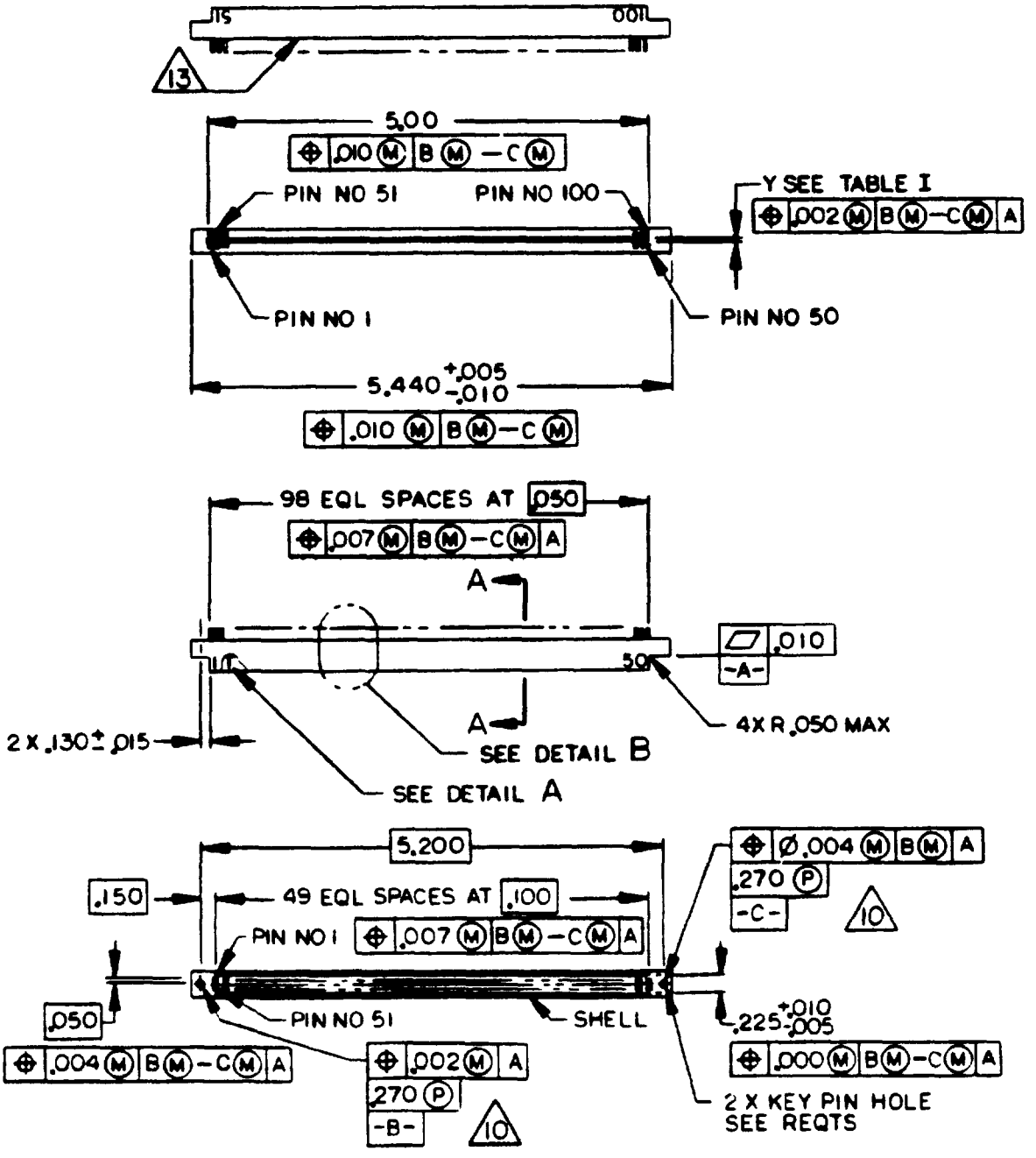
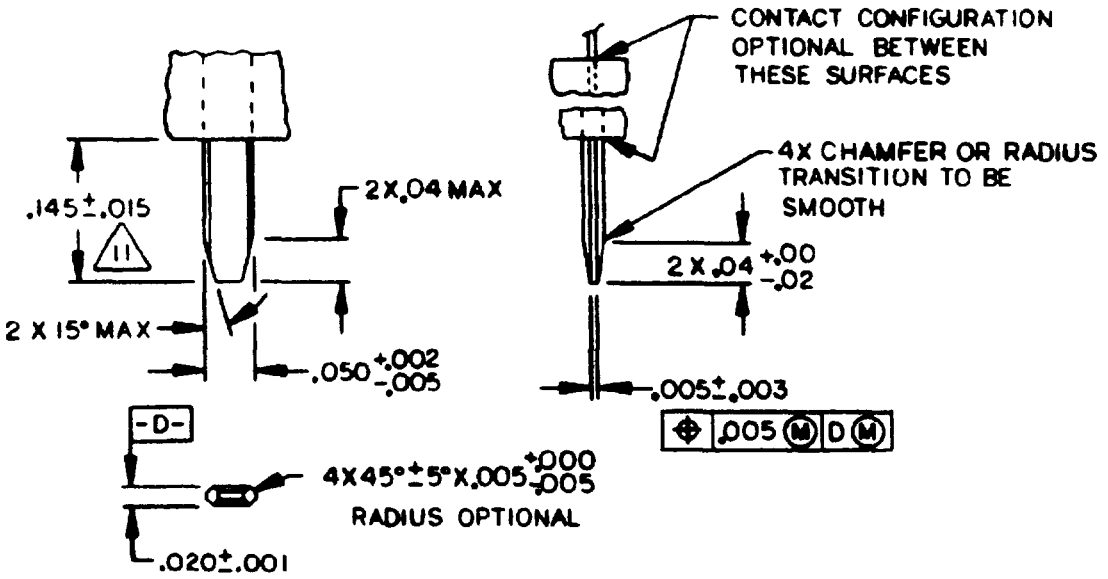


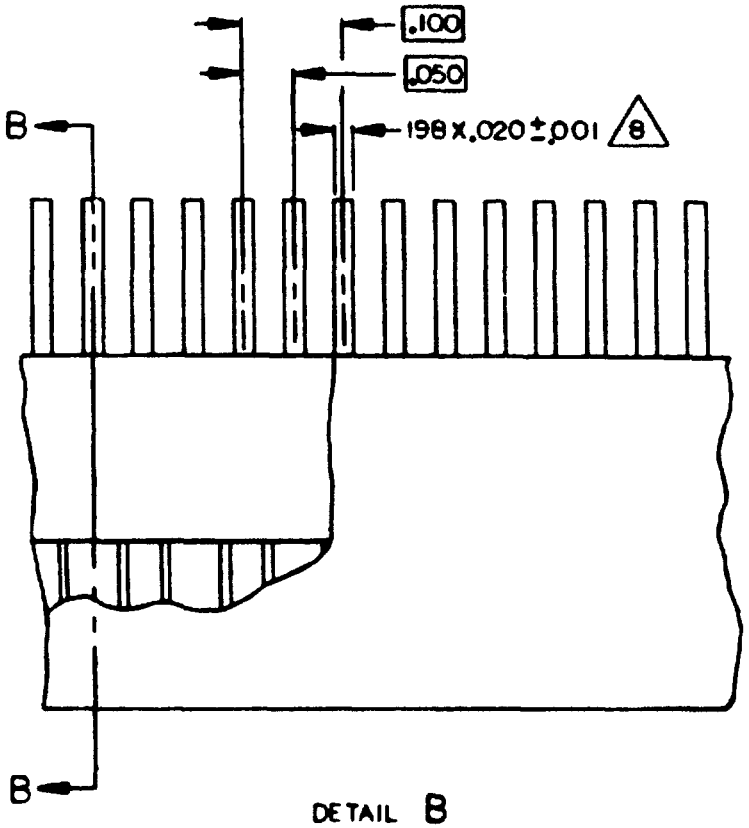
FIGURE 1. Dimensions and configurations

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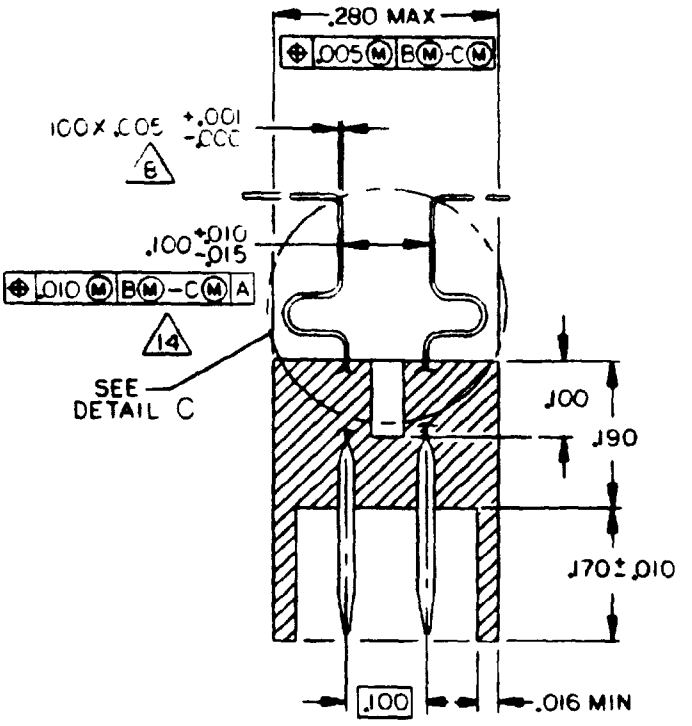
DETAIL A
100 X



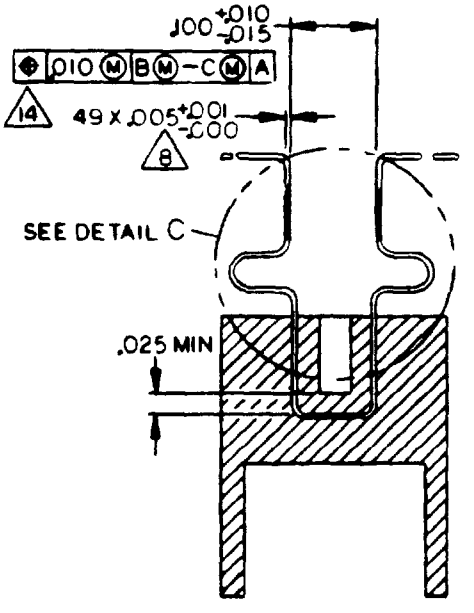
DETAIL B

FIGURE 1. Dimensions and configurations - continued

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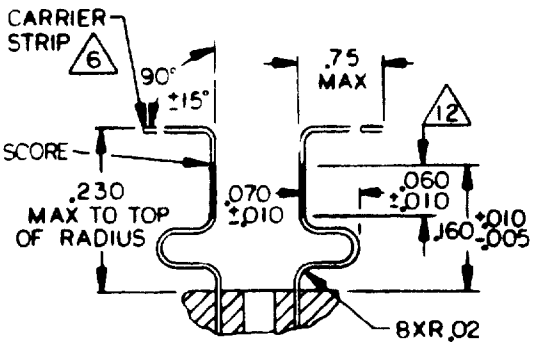
SECTION A-A



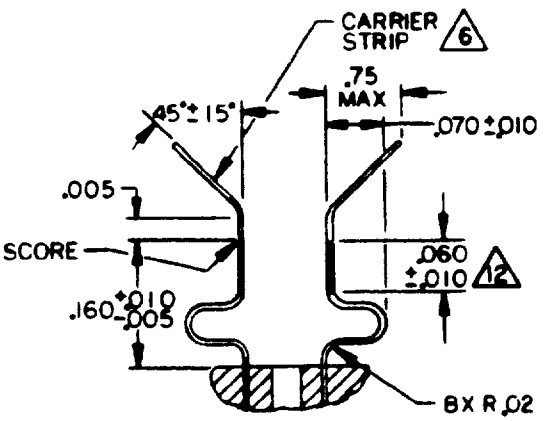
SECTION B-B

FIGURE 1 Dimensions and configurations - continued

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DETAIL C



DETAIL C
CONFIGURATION 1

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
001	03	.015	.38	.070	1.77	.190	4.83	5.200	132.08
002	.05	016	.41	.100	2.54	.225	5.72	5.440	138.18
003	08	.020	51	130	3.30	.230	5.84		
004	10	.025	64	.145	3.68	.270	6.86		
005	13	.040	1 02	150	3.81	.280	7 11		
007	18	050	1.27	160	4.06	.750	19.05		
010	25	060	1 52	170	4.31	5.000	127.00		

FIGURE 1 Dimensions and configurations - continued

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (13 mm) inch for three place decimals, ±.01 (.3 mm) inch for two place decimals and ±2° for angles.
4. Flash around blade contacts not to exceed .010 (25 mm) inch from the surface of the insulator and shall be firmly attached.
5. Pin shield shall be marked with contact numbers 1, 50, 51 and 100 in contrasting color ink .06 (1.5 mm) inch high.
6. The carrier strip shall remain attached to the contact tails to facilitate shipping and may be tinned. It shall be scored at the point indicated to facilitate removal at final assembly. The carrier strip is not required to be continuous.
7. Single or multiple piece construction is optional provided that the end requirements of the drawing are met. It is permissible for the connector body to be installed in either an aluminum or thermoplastic shell which provides the pin shield and key pin hole features The thermoplastic shall be type GLT-30F.
8. Dimension prior to solder coating
9. Dimensions apply after plating.
10. See Table I for key pin hole size and body material.
11. Contact engagement length.
12. Shaded area shall be solder coated.
13. Connector shall be marked in approximate location shown with part number, manufacturer's number and date code with contrasting ink in characters .06 (1.5 mm) inch high
14. This applies only to the .060 ± .010 surfaces
15. Plating defects on the carrier strip above the score line are acceptable

FIGURE 1. Dimensions and configurations - continued

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REQUIREMENTS

Dimensions and configuration See Figure 1 and Table I

Materials Contacts shall be either copper alloy sheet or strip, UNS Number C72500, temper 1/4 hard minimum, in accordance with ANSI/ASTM B 122 or brass, QQ-B-613, alloy 260, temper 1/2 hard.

Connector body shall be either polyetherimide, Type GLT-20F or GLT-30F, molded thermoplastic in accordance with MIL-M-24519 or molding plastic, diallyl phthalate resin, GL filler, MIL-M-14, Type GDI-30F or Type SDG-F Optional end blocks shall be aluminum alloy 6061-T6 per QQ-A-200/8

The pin shield shall be aluminum alloy 5052-H34 per QQ-A-250/8, 6063-T6 per QQ-200/9, or glass-epoxy, GF-0160-00/00-0-1-X, MIL-P-13949 or polyetherimide, Type GLT-20F or GLT-30F, molded thermoplastic in accordance with MIL-M-24519 Pin shield thickness for molded thermoplastic shall be 0 020 (51 mm) inch minimum

Plating Contacts shall be finished on all exposed surfaces The contact engagement length indicated shall be finished with gold plate, MIL-G-45204, Type II, Grade C, Class 1, 0 000050 (0 00013 mm) inch minimum thickness over nickel plating, QQ-N-290, Class 1, 0 000050 (0.00013 mm) inch minimum thickness

Plating requirements for connector solder tails shall be as follows

Steps 1, 2, 4, 5, 7 - Clean prior to plating per MIL-P-81728 MIL-P-81728 states cleanliness requirements prior to plating.

Step 3 - Cu plate per MIL-C-14550 Class 4 to a thickness of 0 0001 (0.0025 mm) inch.

Step 6 - Plate per MIL-P-81728 to a thickness of 0 0003 - 0 0005 (0.0076 - 0 0127 mm) inch Alloy 90% SN-10% Pb

Step 8 - Bake as per Section 3.3 4 of MIL-P-81728

Step 9 - Form

Step 10 - Fuse/reflow with peanut oil as suggested. Quality assurance of MIL-P-81728 shall apply No sign of dewet or poor solderability will be accepted

Step 11 - Avoid chlorinated solvents in rinse.

Step 12 - Selectively solder coat the leads as shown in the shaded area with QQ-S-571, SN60 or SN63 to a thickness of 0.0005 - 0.003 (0.0005 - 0 0800 mm) inch using flux, MIL-F-14256, Type RMA

All aluminum alloy parts shall be anodic coated in accordance with MIL-A-8625, Type III, Class 2, black

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Key pin holes: For M28754/61-05 and -06 material, .095 ± .001 (2.41 ± .03 mm) inch diameter x .150 (3.81 mm) inch deep.

For M28754/61-03 and -04 material, .098 ± 001 (2.48 ± 03 mm) inch dia x .150 (3 81 mm) inch deep.

For M28754/61-01 and -02 material, .114 $^{+.000}_{-.005}$ (2.90 $^{+.00}_{-.13}$ mm) inch dia x .150 (3 81 mm) inch deep

C'bore 134 $^{+.005}_{-.000}$ (3.40 $^{+.13}_{-.00}$ mm) inch dia x .020 $^{+.005}_{-.000}$ (51 $^{+.13}_{-.00}$ mm) inch deep

Three equally spaced 0.010 (.25 mm) inch rad beads x .150 (3 81 mm) inch deep around the periphery of the 114 (2 90 mm) inch diameter shall be utilized

Electrical requirements·

Dielectric withstanding voltage: The connector shall meet the requirements of MIL-STD-1344, Method 3001 for 600 VAC (RMS) 60 Hz, or 850 VDC. The point of measurement shall be between adjacent contacts.

Low level (dry circuit): Voltage drop shall not exceed 36 millivolts. Wire size is not applicable

Contact resistance· Voltage drop shall not exceed 36 millivolts Current Rating; 3 Amp

Temperature rating· -55°C to +125°C

Voltage rating· 300 Volts, AC (RMS), at sea level

Durability: In accordance with MIL-C-28754 and shall not exceed 36 millivolts.

Solderability. In accordance with MIL-C-28754

Military part number M28754/61- (Dash Number from Table I)

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TABLE I			
DASH NUMBER	DIMENSION "Y"	KEY PIN HOLE SIZE INCH (MM)	SHELL MATERIAL
01	060± 002 (1 52± 05)	$114^{+\ 000}_{-\ 005}$ ($2\ 90^{+\ 00}_{-\ .13}$ mm)	Type GDI-30F or SDG-F
02	034±.002 (0.86±.05)	$114^{+\ 000}_{-.005}$ ($2.90^{+\ 00}_{-\ 13}$ mm)	Type GDI-30F or SDG-F
03	.060± 002 (1 52±.05)	098±.001 (2 49±.03)	Aluminum Alloy
04	034± 002 (0 86± 05)	098± 001 (2 49± 03)	Aluminum Alloy
05	.060±.002 (1 52± 05)	.095± 001 (2 41± 03)	Type GLT-20F or GLT-30F
06	034± 002 (0 86± 05)	.095± 001 (2 41± 03)	Type GLT-20F or GLT-30F
CONFIGURATION 1			
07	060± 002 (1 52± 05)	$.114^{+\ 000}_{-\ 005}$ ($2.90^{+\ 00}_{-\ 13}$ mm)	Type GDI-30F or SDG-F
08	034± 002 (0 86± 05)	$.114^{+\ 000}_{-\ 005}$ ($2\ 90^{+\ 00}_{-\ .13}$ mm)	Type GDI-30F or SDG-F
09	.060±.002 (1.52±.05)	.098±.001 (2.49±.03)	Aluminum Alloy
10	.034± 002 (0 86±.05)	098± 001 (2 49± 03)	Aluminum Alloy
11	060±.002 (1 52± 05)	095± 001 (2 41± 03)	Type GLT-20F or GLT-30F
12	034±.002 (0.86±.05)	095± 001 (2.41±.03)	Type GLT-20F or GLT-30F

Warning The GLT-20F or GLT-30F may be susceptible to stress cracking when exposed to certain cleaning solvents.

Qualification information. The connector shall meet the qualification test of MIL-C-28754 Table IV, Type IV, except for Subgroup IV, Method Paragraph 4.7.18, 4.7.19 and 4.7.20. All testing, except visual and mechanical, shall be done with the connector attached to a ceramic substrate as described in Figure.

Number of units to be inspected. Twelve (12) of each dash number to be qualified shall be inspected.

Supersession data: Supersedes Naval Electronic Systems Command Drawing Number 0102-711-1, -2, -4, -5, -7 and -8, Code Identification Number 28687.

Custodians:
Navy - EC

User Activities:
Navy - AS

Preparing Activity:
Navy - EC

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